

Salifort Motors — Employee Retention Project Proposal

ISSUE / PROBLEM

Salifort Motors faces ~17% employee attrition. Finding and hiring replacements is costly. HR needs a data-driven way to identify at-risk employees before they resign.

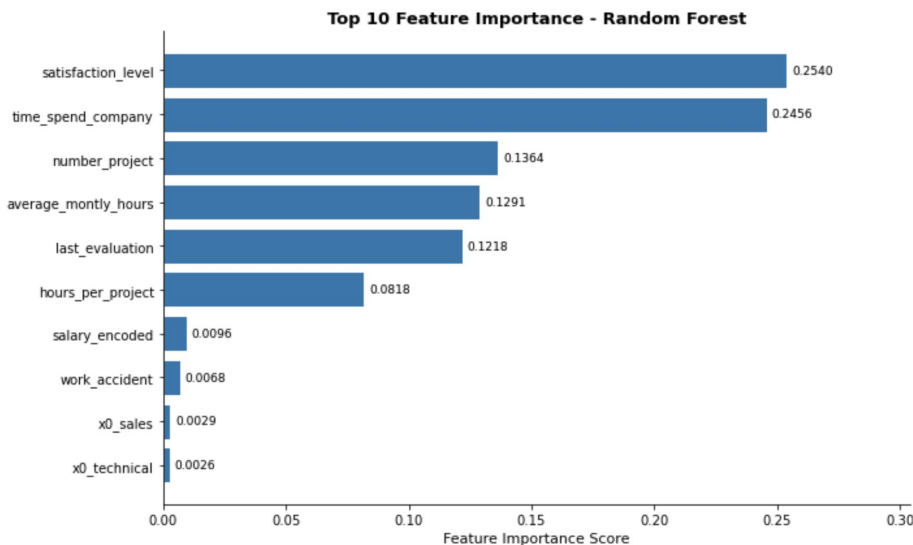
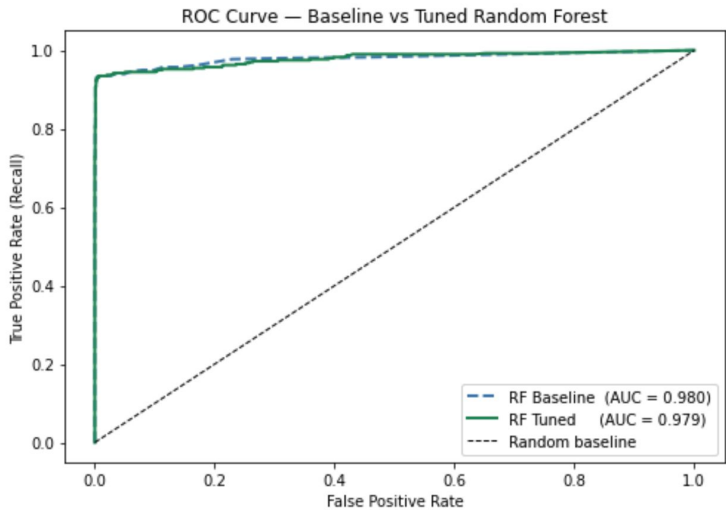
RESPONSE

Analyzed HR data from ~15,000 employees. Built and compared three classification models. Selected a tuned Random Forest as the final model, prioritizing Recall to minimize missed at-risk employees.

IMPACT

Metric	Score
Recall	92.2%
F1 Score	95.3%
ROC-AUC	97.9%

The model correctly flags 9 out of 10 employees likely to leave, enabling proactive HR intervention.



Top attrition drivers identified by the model: **satisfaction level, tenure, average monthly hours, number of projects, and last evaluation score**. Attrition peaks sharply at **year 5 (45.4%)**. Employees with no promotion face 4x higher attrition than those who were promoted.

KEY INSIGHTS

- ➔ **1. Two leaver archetypes drive most attrition:**
- ➔ **Burned Out** — overworked, high performer, no promotion
- ➔ **Disengaged** — underutilized, low engagement, no promotion
- ➔ Attrition peaks at year 5. No promotion = 4x higher attrition risk. Low salary and extreme workload (2 or 7 projects) are the strongest warning signals.
- ➔ **2. Next Step:**
- ➔ Cap projects at 3–5 per employee; flag 250+ monthly hours for review
- ➔ Launch retention conversations at the 3-year tenure mark
- ➔ Link high performance scores to promotion or compensation review
- ➔ Re-run model quarterly as new HR data is collected